Lecture 19, Math 441

Aldo Manfroi

University of Illinois at Urbana-Champaign

Method of Undetermined Coefficients (cont.)

Ex: $y'' - 2y' + y = 6\,\mathrm{e}^{x}$.

$y'' - 2y' + y = 0$. Using $y_c = \mathrm{e}^{rx}$ we get $r^2 - 2r + 1 = 0$.

$y_1 = \mathrm{e}^{x}$; $y_2 = x\mathrm{e}^{x}$

$r = 1$ twice, so $y_c(x) = c_1\,\mathrm{e}^{x} + c_2\,x\,\mathrm{e}^{x}$.

$y_p = A\,x^2\,\mathrm{e}^x.$

$y_p' = \cdots$

$y_p'' = \cdots$

Find A.

$y = y_c + y_p$

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Ex: $y''' - y' = x^2$.

$y_c : \; y''' - y' = 0$

$y_c = \mathrm{e}^{rx} \qquad r^3 - r = 0 \;\rightarrow\; r(r^2 - 1) = 0 \;\rightarrow\; r = 0; r = \pm 1$

$y_c = c_1 + c_2\,\mathrm{e}^{x} + c_3\,\mathrm{e}^{-x}$

$y_p = (A\,x^2 + B\,x + C)x$

$y_p = A\,x^3 + B\,x^2 + Cx$

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Ex: $y^{(4)} - y'' = x + \cos(2x)$

$y_c : \; y^{(4)} - y'' = 0$

$y_c = c_1 + c_2\, x + c_3\, \mathrm{e}^{x} + c_4\, \mathrm{e}^{-x}$

$y_p = (Ax + B)x^2 + (C\cos(2x) + D\sin(2x))$

$y_p = Ax^3 + Bx^2 + C\cos(2x) + D\sin(2x)$

Find A, B, C, D.

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Method of Undetermined Coefficients

Let's amend our rule for this method to find y_p:

If the left hand side is constant coefficient and the right hand side is an exp., sin. or cos., or a polynomial, or a product of them, try a y_p of the same form, add all possible derivatives, each with a coefficient, multiply by x as many times as necessary to avoid duplication with y_c, and then determine the coefficients.

Ex: $y'' + y = \ln(x)$

$y_p = A\ln(x) + \frac{B}{x^2} + \frac{C}{x^4} + \frac{D}{x^6} + \cdots$

$y_p' = \frac{A}{x} - \frac{2B}{x^3} - \frac{4C}{x^5} + \cdots \qquad y_p'' = \frac{-A}{x^2} + \frac{6B}{x^4} + \frac{20C}{x^6} + \cdots$

This would never stop, because $\ln(x)$ does not have a finite number of different derivatives.

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Method of Variation of Parameters

Ex: $y'' + y = \ln(x)$

$y_c : \; y'' + y = 0$

$y_c = c_1 \cos(x) + c_2 \sin(x)$

$y_p(x) = u_1(x) \cos(x) + u_2(x) \sin(x)$

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Now we calculate y_p':

$$y_p' = u_1' \cos(x) - u_1 \sin(x) + u_2' \sin(x) + u_2 \cos(x)$$

$$u_1' \cos(x) + u_2' \sin(x) = 0$$

$$y_p' = \cancel{u_1' \cos(x)} - u_1 \sin(x) + \cancel{u_2' \sin(x)} + u_2 \cos(x)$$

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$$y_p'' = -u_1' \sin(x) - u_1 \cos(x) + u_2' \cos(x) - u_2 \sin(x)$$

Substituting y_p and y_p'' in the original equation ($y'' + y = \ln(x)$):

$$\underbrace{-u_1' \sin(x) - \cancel{u_1 \cos(x)} + u_2' \cos(x) - \cancel{u_2 \sin(x)}}_{y_p''} +$$

$$\underbrace{\cancel{u_1 \cos(x)} + \cancel{u_2 \sin(x)}}_{y_p} = \ln(x)$$

which simplifies to

$$-u_1' \sin(x) + u_2' \cos(x) = \ln(x)$$

$$\begin{cases} -u_1' \sin(x) + u_2' \cos(x) = \ln(x) \\ u_1' \cos(x) + u_2' \sin(x) = 0 \end{cases}$$

$$u_2' = -\frac{\cos(x)}{\sin(x)} u_1'$$

$$-u_1 \sin(x) - \frac{\cos^2(x)}{\sin(x)} u_1' = \ln(x)$$

$$(-\sin^2(x) - \cos^2(x)) u_1' = \sin(x) \ln(x)$$

$$\begin{cases} u_1' = -\sin(x) \ln(x) \\ u_2' = \cos(x) \ln(x) \end{cases}$$

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$$u_1(x) = -\int^x \sin(s)\,\ln(s)\,\mathrm{d}s$$

$$u_2(x) = \int^x \cos(s)\,\ln(s)\,\mathrm{d}s$$

$$y_p = u_1(x)\,\cos(x) + u_2(x)\,\sin(x) =$$

$$-\cos(x)\,\int^x \sin(s)\,\ln(s)\,\mathrm{d}s + \sin(x)\,\int^x \cos(s)\,\ln(s)\,\mathrm{d}s$$

$$y = y_c + y_p$$

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